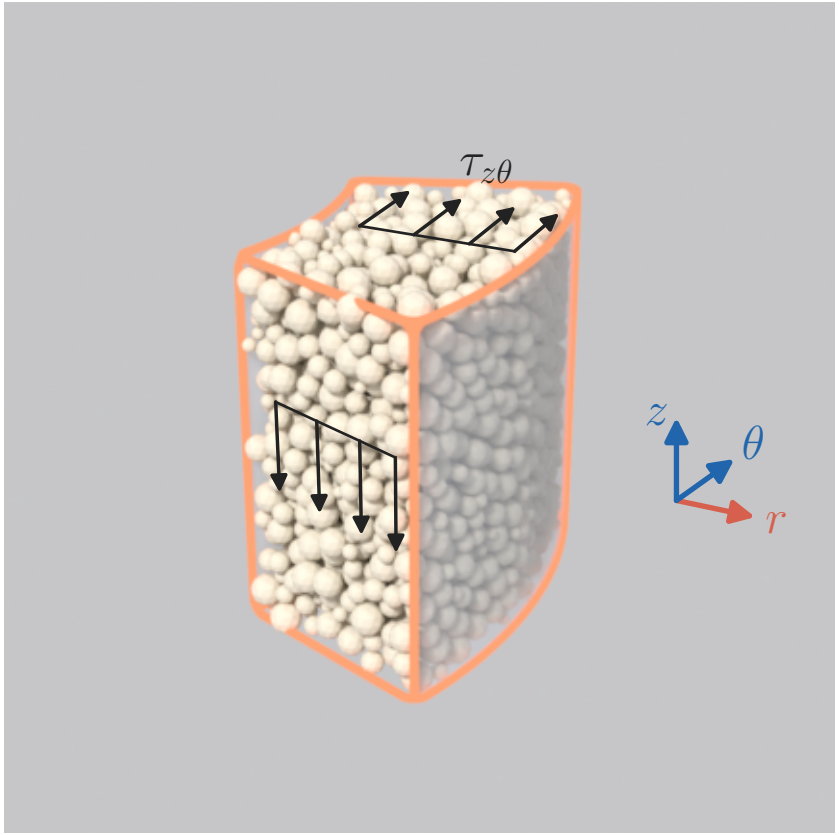

axial (z):

$$H = 100\,\mathrm{mm} \approx 33\,d_{\max}$$

circumferential (θ):

$$\pi(r + R) \approx 251\,\mathrm{mm} \approx 84\,d_{\max}$$

$\tau_{z\theta}$ acts in the (z, θ) plane, orthogonal to the radial direction.